

Simulating Wildfire Mitigation: A Multi-Agent Approach to Community Behavior

Anonymous submission

Appendix Interviewee Demographics

Category	Group	% of Sample ($n=20$)
Compliance Status	Compliant	55.00
	Partially Compliant	36.00
	N/A	9.09
Insurance Status	Retained	55.00
	Non-Renewed	27.00
	N/A	18.00

Table 1: Summary of compliance and insurance status for the 20 interviewed homeowners.

Consent Statement. All 20 homeowners interviewed for this study were informed of the research purpose and voluntarily agreed to participate.

Agent Memory Seed Importance Scoring

Memory Seed Importance Scoring (1–10):

- 9–10 = defining life/property event
- 7–8 = significant action or strong emotional reaction
- 5–6 = meaningful conversation or decision about mitigation
- 3–4 = routine observation or minor step
- 1–2 = passing remark or peripheral detail

Simulation Judge Prompts

Each judgment scores a single dimension in an independent call. Behavioral plausibility and intervention responsiveness are shown the resident’s objective circumstances but not the seed narrative; persona consistency is shown the seed narrative and memory seeds but not the simulation events. The judge (openai/gpt-5.4, temperature 0) writes its reasoning before assigning an integer score from 1 to 5 and returns JSON.

Behavioral Plausibility

You are evaluating ONE dimension: behavioral plausibility. You are shown this resident’s objective circumstances, but not their personality,

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attitudes, or history. Judge whether the reasoning makes sense for someone in these circumstances. Do not judge whether it sounds like a particular person .

Judge the resident’s REASONING. The decision is shown as context: you are assessing whether the reasoning that produced it makes sense for someone in these circumstances.

BEHAVIORAL PLAUSIBILITY: How plausible, if at all, is this reasoning for a homeowner facing this situation ?

- 1 = Not plausible. Incoherent, contradicts basic facts of the situation, or no homeowner would reason this way.
- 2 = Barely plausible. The reasoning is followable but rests on a clear misunderstanding of the situation or makes a leap no homeowner would make.
- 3 = Reasonable but generic. Sensible, but the reasoning could apply to almost anyone facing this situation.
- 4 = Reasonable and situationally specific. The reasoning engages concrete features of this situation (costs, timing, constraints).
- 5 = Nuanced situational understanding. Weighs competing pressures or tradeoffs specific to this situation in a way that shows a meaningful grasp of what is at stake.

Rate on an integer scale of 1-5. Use whole numbers only. Write your reasoning first, then your score.

Persona Consistency

You are an expert evaluator assessing resident decisions in a wildfire mitigation study. Read the seed narrative carefully to understand this resident’s actual role and constraints before scoring.

You are evaluating ONE dimension: persona consistency. You are shown the resident’s seed narrative and memory seeds, but not the events of the simulation. Judge only faithfulness to the seed.

PERSONA CONSISTENCY: How different, if at all, would this response be if the agent had a different seed narrative? Note: distinctive communication style, tone, and framing count as persona signals.

1 = Contradicts the seed personality or key memory seeds
.
2 = No contradiction, but off. Nothing conflicts outright, but tone, priorities, or framing don't fit this agent.
3 = Consistent with a generic version of this persona type but lacking in nuance, for example, the agent's specific history, experiences, or distinctive voice
.
4 = Reflects at least one detail unique to this agent, for example, a specific cost, person, place, opinion, characteristic tone, or prior experience from the seed narrative or memory seeds.
5 = Clearly reflects details unique to this agent: specific costs, people or places, distinctive opinions, characteristic tone, or prior experiences from the seed narrative or memory seeds.

Rate on an integer scale of 1-5. Use whole numbers only.
Write your reasoning first, then your score.

Intervention Responsiveness

You are an expert evaluator assessing resident decisions in a wildfire mitigation study.

You are evaluating ONE dimension: intervention responsiveness. You are shown this resident's objective circumstances but not their personality, attitudes, or history. Judge only how specifically the response engages the content of the intervention
.

INTERVENTION RESPONSIVENESS: How specifically, if at all, did the agent engage with the content of this intervention?

1 = Ignored the intervention or gave a fully generic response.
2 = Acknowledged the intervention happened, but the response could apply to multiple different interventions.
3 = Acknowledged the event and responded appropriately but did not engage with its specific details.
4 = Engaged with specifics. Referenced concrete details from this intervention, for example, deadlines, requirements, named parties, or amounts.
5 = Acted on specifics. The response turned on particular details of this intervention; a different deadline, amount, or requirement would have produced a different decision.

Rate on an integer scale of 1-5. Use whole numbers only.
Write your reasoning first, then your score.

All three prompts instruct the judge to output JSON only, as a dictionary with keys reasoning (a string) and score (an integer from 1 to 5).

Detailed Agent Tuning and Evaluation

Parameter	Definition
Top-K	How many memories to pass to agent prompt at decision time (5/8/12)
Recency Weight (α)	How much to weight how recently a memory was created. Memories from earlier simulation days score lower using exponential decay
Importance Score (β)	How much to weight the importance score that a memory has
Relevance Weight (γ)	How much to weight the semantic similarity between the intervention and each memory
Retrieval Mode	Which algorithm computes the semantic similarity between intervention and each memory. Dense: HuggingFace embeddings + cosine similarity; Sparse: keyword matching; Hybrid: blend of both, controlled by sparse_weight

Table 2: Retrieval hyperparameter definitions.

Top-K	Recency Weight	Importance Weight	Relevance Weight	Retrieval Mode
8	1.0	1.0	1.0	Dense
5	1.0	1.0	1.0	Dense
12	1.0	1.0	1.0	Dense
8	0.5	1.0	2.0	Dense
8	0.5	1.0	3.0	Dense
8	1.0	2.0	1.0	Dense
8	2.0	1.0	1.0	Dense
8	1.0	1.0	1.0	Sparse
8	1.0	1.0	1.0	Hybrid

Table 3: Retrieval configurations tested.

Memory Retrieval Hyperparameter Tuning and Evaluation:

Configuration	Insight	Accuracy	Combined
Threshold=50, questions=3	7/10	7/10	7/10
Threshold=50, questions=5	7/10	6/10	6.5/10
Threshold=100, questions=3	7/10	6/10	6.5/10
Threshold=150, questions=3	6/10	6/10	6.0/10

Table 4: Reflection configurations tested.

Reflection Hyperparameter Tuning and Evaluation:

Detailed Pointwise Scoring Results

These tables give the full per-agent, per-criterion, and per-intervention breakdown behind the pointwise scores summarized in the main text. Because the simulation agents run on Claude models, all scores come from an independent judge from a different model family (openai/gpt-5.4, temperature 0), which avoids self-preference bias. Every agent decision is scored on Behavioral Plausibility (BP), Persona Consistency (PC), and Intervention Responsiveness (IR), and the entire scoring pass is repeated five times and averaged, yielding 1,800 judgments in total.

Overall Scores by Agent and Condition

Condition	Laura	Linda	Walter	Margaret	Miriam Voss
Baseline	4.62	4.51	4.38	4.46	4.67
Ablation 1 (No Reflection)	4.63	4.47	4.47	4.49	4.63
Ablation 2 (No Memory or Reflection)	4.51	4.51	4.60	4.47	4.63
Budget	4.53	4.49	4.54	4.53	4.46

Table 5: Mean of BP, PC, and IR for each agent, averaged over the six interventions and five scoring passes. Indicates whether ablation effects are consistent across agents or driven by specific profiles.

Baseline: Scoring by Criterion and Agent

Criterion	Laura	Linda	Walter	Margaret	Miriam Voss
Behavioral Plausibility	4.67	4.37	4.37	4.27	4.37
Persona Consistency	4.47	4.37	4.30	4.40	4.87
Intervention Responsiveness	4.73	4.80	4.47	4.70	4.77
Mean	4.62	4.51	4.38	4.46	4.67

Table 6: Baseline condition: scoring by criterion and agent, from the independent per-intervention judge (GPT-5.4, five passes averaged). Each value is averaged over the agent’s six interventions.

Baseline Variant: Scoring by Intervention and Agent

Behavioral Plausibility					
Intervention	Laura	Linda	Walter	Margaret	Miriam Voss
Day 1: Firewise ordinance notification	4.60	4.20	4.20	4.00	4.00
Day 12: Defensible space inspection	4.80	4.80	4.00	4.20	4.80
Day 24: Neighbor pressure	4.60	4.60	4.40	4.40	4.40
Day 36: Firewise community campaign	4.20	4.00	4.40	4.00	4.20
Day 48: Insurance non-renewal notice	4.80	4.40	5.00	4.80	4.80
Day 55: Resource assistance email	5.00	4.20	4.20	4.20	4.00
Mean	4.67	4.37	4.37	4.27	4.37

Table 7: Behavioral Plausibility by intervention and agent, Baseline condition (five scoring passes averaged).

Persona Consistency					
Intervention	Laura	Linda	Walter	Margaret	Miriam Voss
Day 1: Firewise ordinance notification	4.40	4.60	5.00	4.60	5.00
Day 12: Defensible space inspection	4.60	4.40	4.20	4.60	5.00
Day 24: Neighbor pressure	4.20	3.80	3.80	4.00	4.40
Day 36: Firewise community campaign	4.60	4.40	4.00	4.40	5.00
Day 48: Insurance non-renewal notice	4.20	4.60	4.40	4.00	5.00
Day 55: Resource assistance email	4.80	4.40	4.40	4.80	4.80
Mean	4.47	4.37	4.30	4.40	4.87

Table 8: Persona Consistency by intervention and agent, Baseline condition (five scoring passes averaged).

Intervention Responsiveness					
Intervention	Laura	Linda	Walter	Margaret	Miriam Voss
Day 1: Firewise ordinance notification	4.00	4.80	4.60	5.00	5.00
Day 12: Defensible space inspection	5.00	5.00	4.80	5.00	5.00
Day 24: Neighbor pressure	4.80	4.80	4.00	4.60	5.00
Day 36: Firewise community campaign	5.00	4.40	4.20	4.00	4.60
Day 48: Insurance non-renewal notice	5.00	5.00	5.00	5.00	4.40
Day 55: Resource assistance email	4.60	4.80	4.20	4.60	4.60
Mean	4.73	4.80	4.47	4.70	4.77

Table 9: Intervention Responsiveness by intervention and agent, Baseline condition (five scoring passes averaged).

Code Availability

The code is available at <https://anonymous.4open.science/r/wildfire-agent-simulation/README.md>.